

Krishnaa Vadivel | Machine Learning Engineer | High-Performance Computing

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Profile

PhD researcher and engineer building machine-learning models and running them on high-performance computing systems for large-scale materials simulation. Combines model development, GPU-aware implementation, distributed compute, and scientific domain knowledge across materials, optics, and agentic ML tooling.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Developed equivariant graph neural networks in PyTorch and PyTorch Geometric, similar to DeepH and MACE, to accelerate optical-properties calculations and material-energy prediction with automatic differentiation.
- Built distributed compute and data-processing software in Python and C++ using MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc across national HPC environments including Frontier, Frontera, and Perlmutter.
- Connected ML methods to domain theory through first-principles exciton calculations, many-body modeling, and numerical simulation, enabling realistic ML problem framing for materials systems.
- Used agentic ML and retrieval-driven tooling to generate first-principles software inputs from journal articles and technical documentation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state problems.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built CUDA-accelerated data-processing and visualization software for large waveform datasets, strengthening experience in GPU-enabled high-throughput pipelines.
- Built database- and web-backed data-management systems with ASP-style services and Microsoft SQL for collecting, organizing, and processing large-scale well-tool datasets.
- Worked close to embedded and instrumentation contexts, improving practical understanding of data generation, system integration, and performance constraints.

Selected Projects

Materials ML for Optical Properties

- Developed equivariant graph neural networks in PyTorch and torch_geometric to accelerate optical-properties calculations and material-energy prediction in materials systems.

Matrix-Multiply AI Accelerator

- Built a Verilog-based matrix-multiply accelerator project to demonstrate hardware-aware AI compute and datapath design fundamentals.

Agentic RAG for Scientific Workflows

- Built retrieval-driven tooling that reads software documentation and academic papers to generate structured starter inputs for computational science and quantum-chemistry-style tasks.

Container Orchestration for First-Principles Computing

- Developed CUDA-ready and cross-environment containers for first-principles software stacks and published them to Docker Hub for portable scientific computing.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing work spanning scientific computing, first-principles materials simulation, and ML-assisted acceleration.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Selected Coursework

Machine Learning: Probabilistic Machine Learning

Theory: Theoretical Computer Science; strong mathematical/theory preparation supporting modern ML work

Scientific Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Engineering Foundations: Signals and systems, embedded systems, and numerical/scientific computing practice

Technical Skills

ML: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks, transformers, automatic differentiation

Languages: Python, C++, CUDA, JavaScript, Bash, PowerShell

Scientific Python: NumPy, pandas, SciPy, SymPy, matplotlib

Compute: MPI, OpenMP, CUDA, ROCm, PETSc, SLEPc, mpi4py, petsc4py, slepc4py

Platforms: Linux, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Frontera, Perlmutter